

Understanding and Forecasting Household Electricity Consumption

A Machine Learning Approach to Power Usage Analysis and Forecasting

Dersim Kaya

Greenbootcamps

June 2025

Table of Contents

- 1 Professional Background
- 2 Motivation & Stakeholder
 - Motivation
 - Stakeholder Information
 - Data Introduction
- 3 Data Analysis
- 4 Machine Learning Models and Evaluation
 - ML Models
 - Hyperparameter Tuning Results
 - Feature Importance
 - Visual Evaluation of Model Predictions
- 5 Recommendations to Stakeholder

Professional Background

- Master's degree in Mathematics with experience as a mathematics teacher and research assistant
- Completed professional training in Data Analytics, SQL, Python, and Data Engineering
- Using mathematical theory and AI to develop solutions for complex data analysis and predictive modeling
- [linkedin.com/dersimkaya](https://www.linkedin.com/in/dersimkaya)

Motivation

- Efficient energy use is key to reducing costs and supporting sustainability.
- Forecasting household electricity consumption helps optimize power management.
- This project uses machine learning techniques to analyze and predict power usage.

Stakeholder

- Stakeholder: GreenCity – an energy provider focused on sustainable power supply.
- Goal: Improve demand forecasting for better energy distribution and customer service.
- Interest: Accurate prediction of household electricity consumption to optimize grid management.
- Use case: Enable smarter energy planning and reduce waste.

Data Introduction

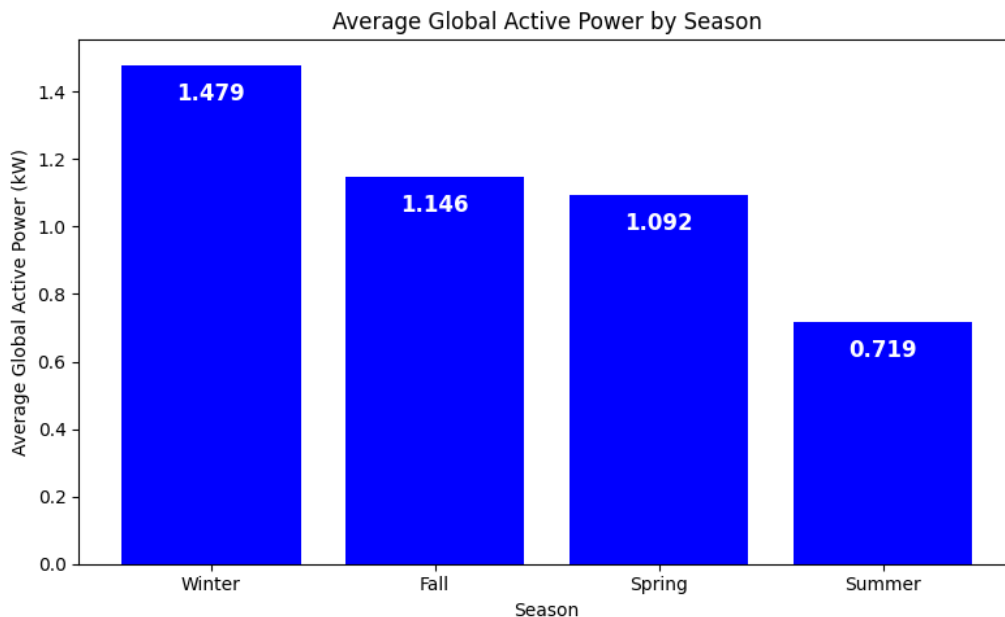
- Dataset contains individual household electric power consumption
- Time series data recorded at one-minute intervals for more than two years.
- Key variables include:
 - Global active power (kW)
 - Sub-metering values (Wh) for kitchen, laundry room, and water heater/AC
 - Date and time stamps
- Data allows exploration of power usage patterns and their changes over time

Data Preprocessing

- Found missing data concentrated in a short time window during the last three days of April 2007

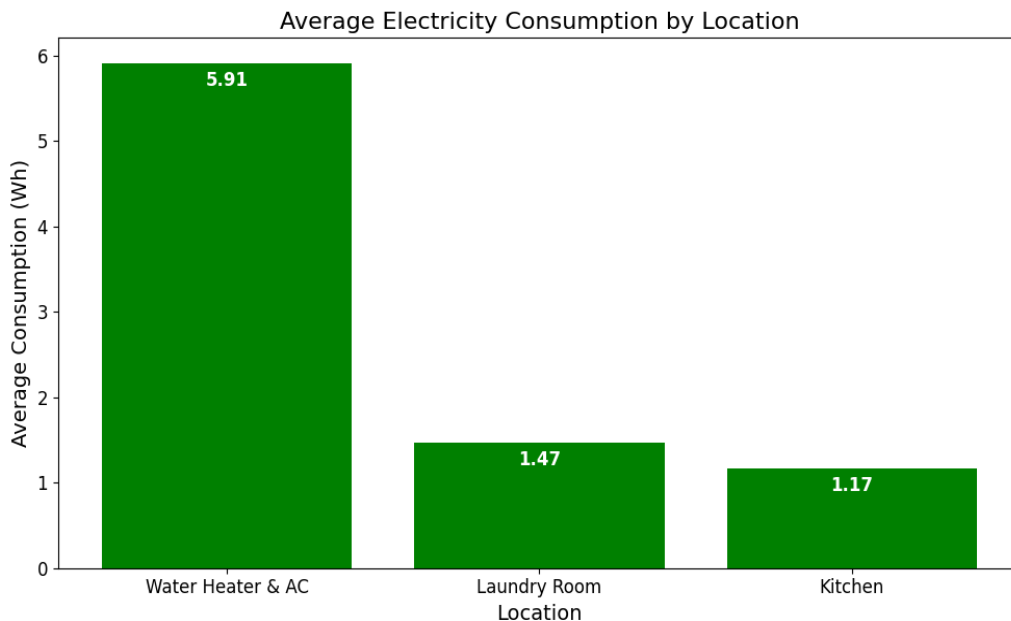
Exploratory Data Analysis (EDA)

H_1 : There is a significant variation in average global active power consumption across different seasons.



Exploratory Data Analysis (EDA)

H_1 : There is a significant difference in average electricity consumption among the Kitchen, Laundry Room, and Water Heater & AC.



Machine Learning Models and Evaluation

Model	RMSE
Gradient Boosting (Test)	0.515
CatBoost Regression (Test)	0.515
XGBoost Regression (Test)	0.515
Random Forest (Test)	0.516
Decision Tree Regression (Test)	0.516
LSTM (Test)	0.517
Polynomial Regression (degree=2)	0.526
Linear Regression (Test)	0.528
LinearSVR (Test)	0.586

Note: Models are ordered by RMSE (Root Mean Squared Error). Lower RMSE indicates better predictive performance.

Extended Models with Weekday and Lag Features

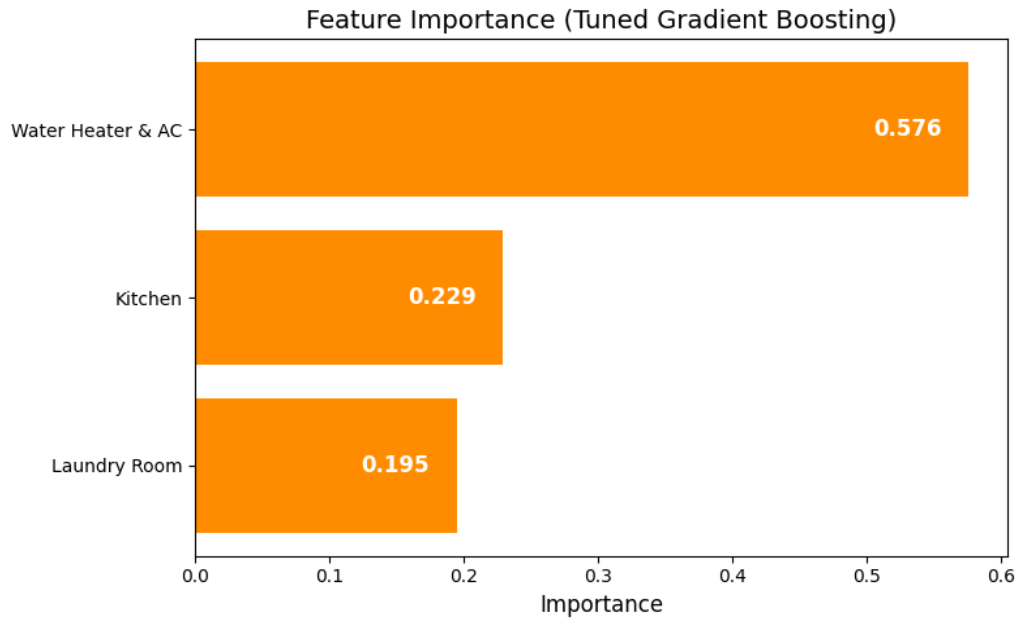
Model	RMSE
Gradient Boosting (Test)	0.515
Gradient Boosting + Weekday (Test)	0.515
Random Forest (Test)	0.516
Lagged LSTM (Test)	0.527
Linear Regression + Weekday (Test)	0.527

Hyperparameter Tuning Results

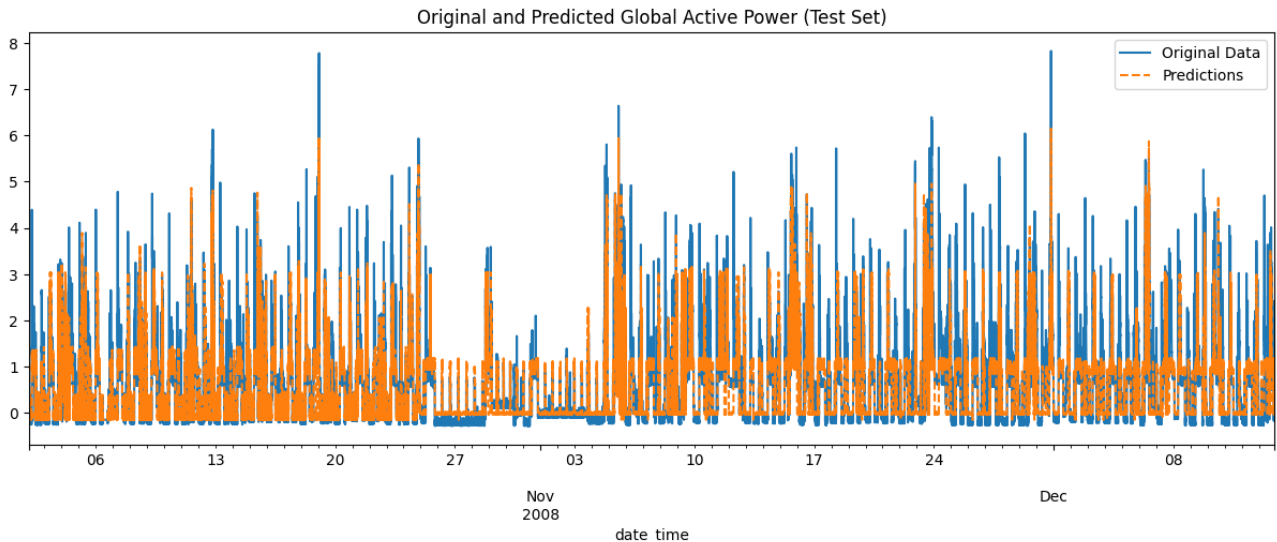
Model	RMSE
Tuned Random Forest (Test)	0.515
Tuned Gradient Boosting (Test)	0.515

Both models achieved identical performance after tuning.

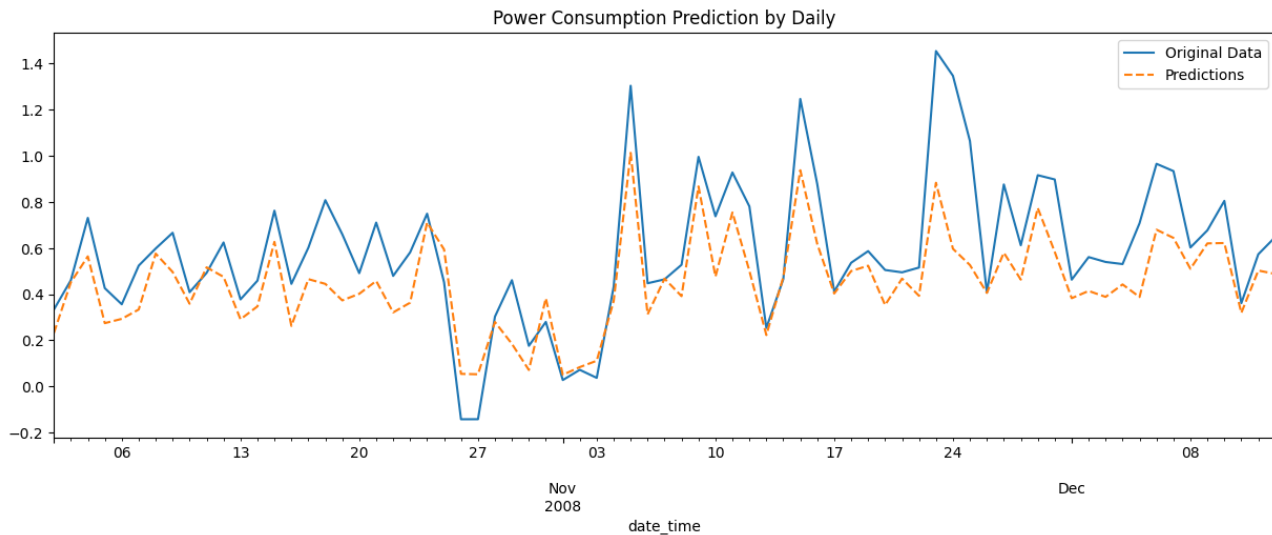
Feature Importance



Original and Predicted Global Active Power



Daily Power Consumption Prediction



Recommendations to Stakeholder

- Investigate data recording issues, especially the missing values at the end of April 2007, to improve data quality.
- Use the Gradient Boosting model for forecasting due to its strong predictive performance.
- Promote the use of energy-efficient appliances, especially for high-consumption devices like water heaters and air conditioning units.
- Leverage model insights to optimize energy distribution and reduce waste.
- Consider seasonal variations in power consumption for better demand planning.

Thank you for your attention!